



9729/01

18 September 2017

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

Multiple Choice Answer Sheet

Write your name, PDG and NRIC / FIN number, **including** the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 PRELIMExam Details: H2 Chemistry / Paper 1

Date: 18/09/2017

This document consists of **16** printed pages.

- 1 Use of the Data Booklet is relevant to this question.

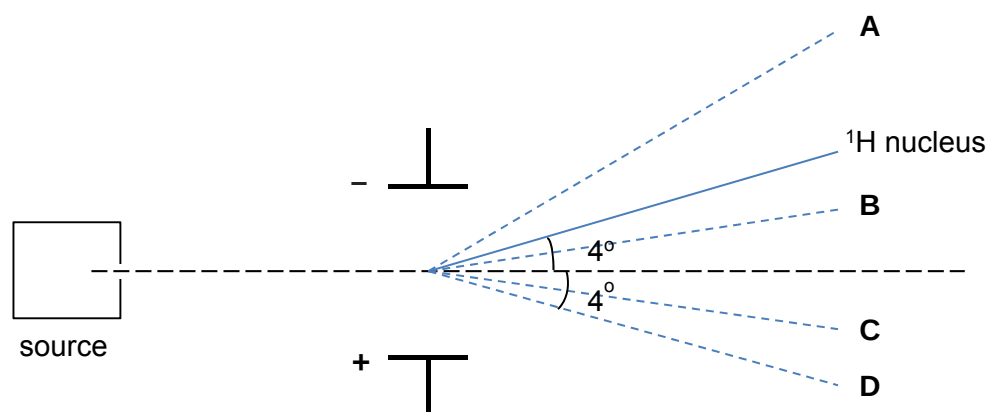
Tritium, ${}^3_1\text{H}$, a radioactive isotope of hydrogen, slowly turns into a helium isotope ${}^3_2\text{He}$.

Which statements about the two isotopes are **incorrect**?

- 1 Both isotopes have more neutrons than electrons.
- 2 Both isotopes have the same number of protons in their nuclei.
- 3 Both isotopes have the same number of charged sub-atomic particles.

- A 2 only
 B 1 and 3 only
 C 2 and 3 only
 D 1, 2 and 3

- 2 When passed through an electric field, the ${}^1\text{H}$ nucleus is deflected as shown below.

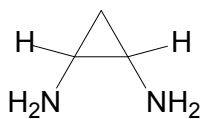
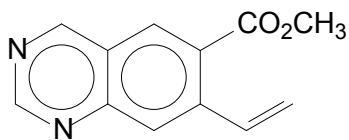
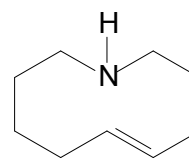


Which of the above beams represents the deflection for an ion ${}^2\text{X}^{2-}$?

- 3 Which of the following molecules / ions has the smallest bond angle?

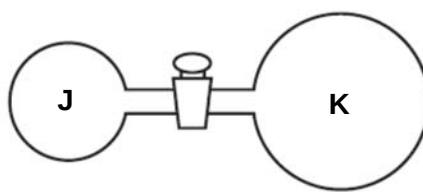
- A SF_2 B XeF_2 C SO_3^{2-} D AlH_4^-

- 4 Compounds **F**, **G** and **H** are cyclic compounds.

**F****G****H**

Which bond is **not** present in the compounds above?

- A** a σ bond formed by sp^3-sp^3 overlap between two C atoms in **F**
 - B** a σ bond formed by sp^2-sp^2 overlap between C and N atoms in **G**
 - C** a σ bond formed by sp^3-sp^2 overlap between two C atoms in **H**
 - D** a σ bond formed by sp^2-sp^2 overlap between C and N atoms in **H**
- 5 Two glass vessels **J** and **K** are connected by a closed valve.

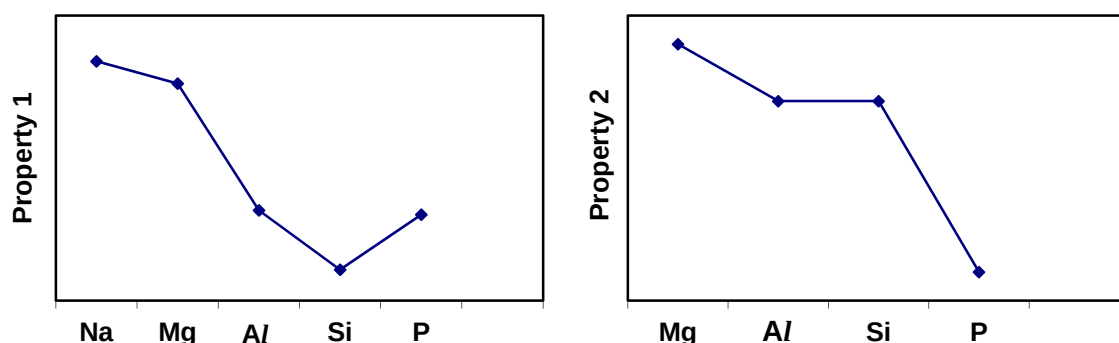


J contains helium gas at 20 °C and a pressure of 1×10^5 Pa. **K** has been evacuated, and has three times the volume of **J**. In an experiment, the valve is opened and the temperature of the whole apparatus is raised to 100 °C.

What is the final pressure in the system?

- A** 3.18×10^4 Pa
- B** 4.24×10^4 Pa
- C** 1.25×10^5 Pa
- D** 5.09×10^5 Pa

- 6 The graphs below show the variation of two properties of some period 3 elements and/or their compounds.



Which of the following correctly describes properties 1 and 2?

	Property 1	Property 2
A	atomic radius of the elements	electrical conductivity of the elements
B	melting point of the chlorides	pH of oxides when added to water
C	melting point of the elements	first ionisation energies of the elements
D	electrical conductivity of elements	pH of chlorides when added to water

- 7 Use of the Data Booklet is relevant to this question.

If iron is heated separately with chlorine, bromine and iodine, what are the likely products?

	chlorine	bromine	iodine
A	FeCl_2	FeBr_2	FeI_2
B	FeCl_3	FeBr_2	FeI_2
C	FeCl_3	FeBr_3	FeI_2
D	FeCl_3	FeBr_3	FeI_3

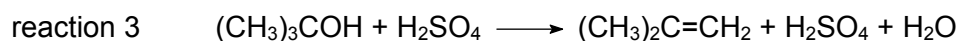
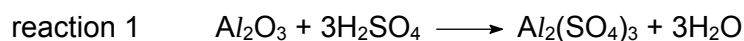
- 8 When 10 cm^3 of a hydrocarbon was burnt in 100 cm^3 of excess oxygen, the volume of the residual gas mixture contracted by 20 cm^3 after passing through a solution of aqueous sodium hydroxide. The remaining volume of gas was just sufficient to completely burn exactly 30 cm^3 of the same hydrocarbon. All volumes were measured at room temperature and pressure conditions.

What is the formula of the hydrocarbon?

- A C_2H_2 B C_2H_3 C C_2H_4 D C_2H_6

- 9 Sulfuric acid, one of the most important industrial chemicals, can carry out several functions in chemical reactions.

Three examples of industrial reactions in which sulfuric acid is used are shown below.

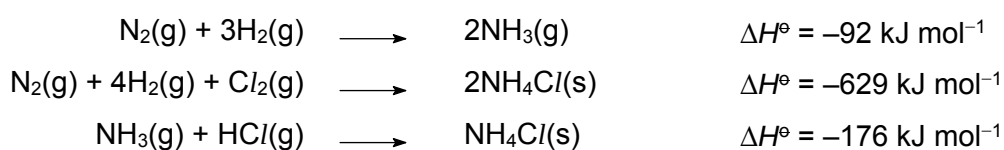


What is the function of sulfuric acid in each reaction?

	reaction 1	reaction 2	reaction 3
A	acidic	oxidising	dehydrating
B	acidic	acidic	dehydrating
C	dehydrating	oxidising	catalytic
D	dehydrating	acidic	catalytic

- 10 Some of these data are also relevant to question 14.

Given the following enthalpy change



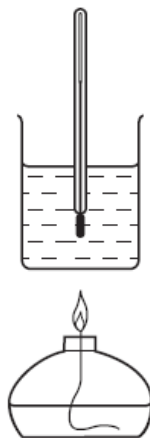
What is the standard enthalpy change of formation of gaseous hydrogen chloride?

- A** -361 kJ mol^{-1}
- B** -185 kJ mol^{-1}
- C** $-92.5 \text{ kJ mol}^{-1}$
- D** $-46.3 \text{ kJ mol}^{-1}$

11 *Use of the Data Booklet is relevant to this question.*

A student carried out an experiment to determine the enthalpy change of combustion of ethanol.

The following results were obtained by the student.



initial temperature of the water	25 °C
final temperature of the water	68 °C
mass of alcohol burner before burning	260.65 g
mass of alcohol burner after burning	259.65 g
mass of glass beaker plus water	160.00 g
mass of glass beaker	60.00 g

Given that the enthalpy change of combustion of ethanol is $-1370 \text{ kJ mol}^{-1}$, what is the efficiency of heat transferred to the water?

A 46%

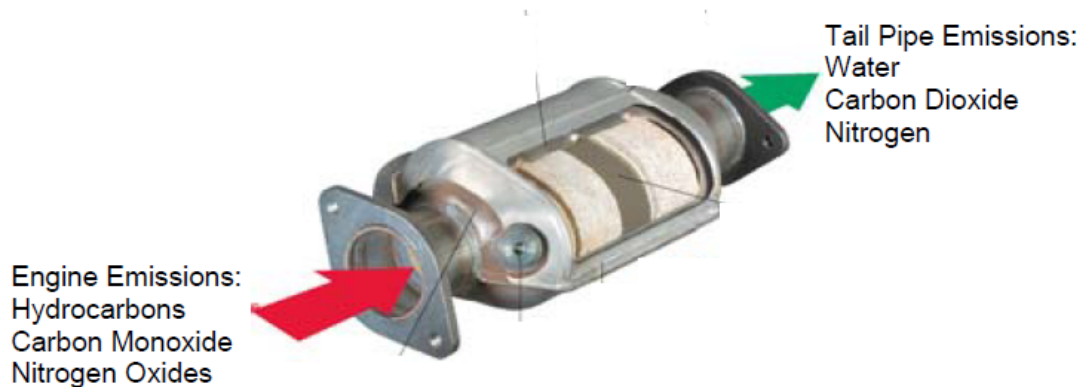
B 60%

C 81%

D 97%

- 12 An example of a three-way honeycomb type catalytic converter is shown below.

The catalytic converter involves three simultaneous reactions. In the first, NO_x gases get reduced; in the second and third, unburnt hydrocarbon and carbon monoxide are completely oxidised.



Which of the following equations does **not** represent the reactions that took place in the catalytic converter?

- A $\text{C(g)} + \text{O}_2\text{(g)} \longrightarrow \text{CO}_2\text{(g)}$
 B $2\text{CO(g)} + \text{O}_2\text{(g)} \longrightarrow 2\text{CO}_2\text{(g)}$
 C $2\text{NO(g)} + 2\text{CO(g)} \longrightarrow \text{N}_2\text{(g)} + 2\text{CO}_2\text{(g)}$
 D $2\text{C}_8\text{H}_{18}\text{(g)} + 25\text{O}_2\text{(g)} \longrightarrow 16\text{CO}_2\text{(g)} + 18\text{H}_2\text{O(g)}$
- 13 A radioactive element has two isotopes, **L** and **M**, with half-lives of 3 min and 9 min respectively.

An experiment starts with n times as many atoms of **L** as of **M**. After 9 min, the number of atoms of **L** and **M** are both equal.

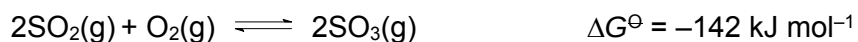
Given that radioactive decay is a first order reaction, what is the value of n ?

- A 0.5 B 2 C 4 D 8
- 14 Use of the data in question 10 is relevant to this question.

Which of the following statement is true about the Haber process for the manufacture of ammonia?

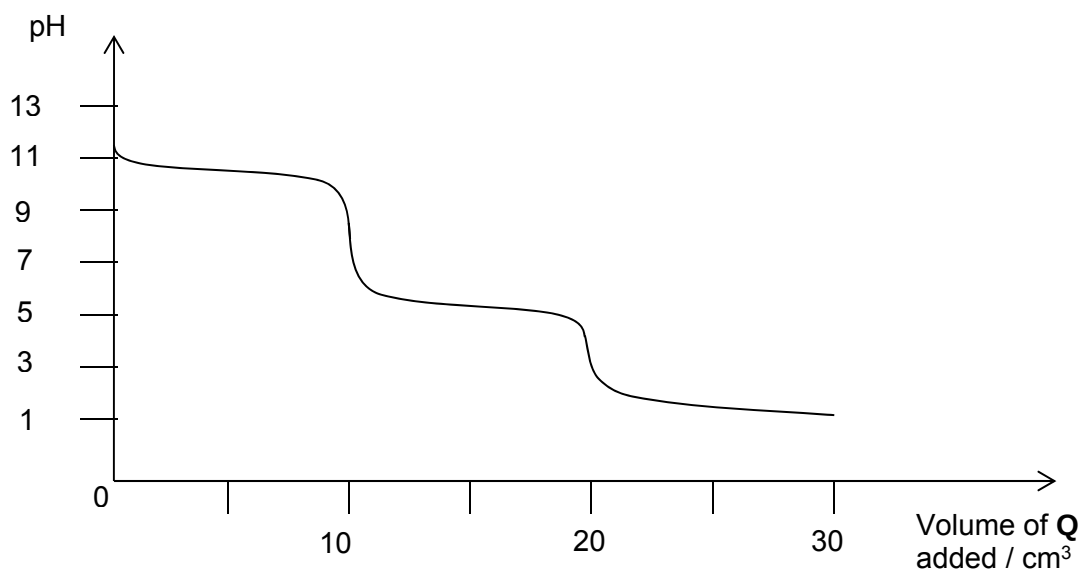
- A Adding a catalyst increases the equilibrium yield of ammonia but the rate of production of ammonia is faster.
 B Adding neon gas at constant pressure increases the equilibrium yield of ammonia but the rate of production of ammonia is slower.
 C Decreasing the temperature increases the equilibrium yield of ammonia but the rate of production of ammonia is slower.
 D Removing nitrogen gas at constant volume increase the equilibrium yield of ammonia but the rate of production of ammonia is slower.

- 15 The Gibbs free energy change of reaction, $\Delta G^\ominus$, for the contact process is shown below.



Which statement about the equilibrium constant, K_p at 25 °C is true?

- A K_p is less than 1.
 B K_p is greater than 1.
 C K_p is equal to 1.
 D No conclusion can be drawn about K_p .
- 16 What is the pH of the final solution formed when 30 cm³ of 0.10 mol dm⁻³ HNO₃ is added to 20 cm³ of 0.20 mol dm⁻³ ammonia?
 [K_b of ammonia = 1.8×10^{-5} mol dm⁻³]
- A 4.3 B 5.2 C 8.8 D 10.8
- 17 The diagram below shows the change of pH produced by gradually adding aqueous **Q** to a certain volume of aqueous **P**. The concentration of each constituent of the aqueous solutions **P** and **Q** is 0.1 mol dm⁻³.



What could **P** and **Q** be?

	P	Q
A	NaOH + KOH	HCl
B	H ₂ NCH ₂ CH ₂ NH ₂	HCl
C	Na ₂ CO ₃	CH ₂ (COOH) ₂
D	KOH	CH ₃ COOH

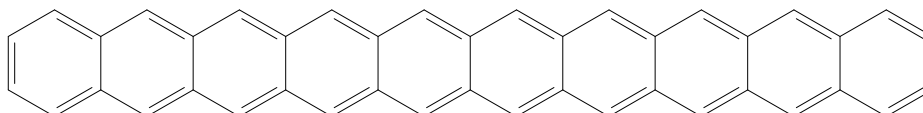
- 18 When heated with chlorine, the hydrocarbon 2-methylbutane, $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)_2$, undergoes free radical substitution.

In a propagation step, which free radical will be formed most readily?

- A $\dot{\text{C}}\text{H}_2\text{CH}_2\text{CH}(\text{CH}_3)_2$
 B $\text{CH}_3\dot{\text{C}}\text{HCH}(\text{CH}_3)_2$
 C $\text{CH}_3\text{CH}_2\dot{\text{C}}(\text{CH}_3)_2$
 D $\text{CH}_3\text{CH}_2\text{CH}(\dot{\text{C}}\text{H}_2)\text{CH}_3$

- 19 Use of the Data Booklet is relevant to this question.

The longest acene, decacene, comprising of 10 linearly fused benzene ring, has been made for the first time in 2017.



Which statements about the molecule are correct?

- 1 Each of the C–C–C bond angle is 120° .
 - 2 The π electrons are delocalised.
 - 3 The bond energy of each carbon–carbon bond in the molecule is equal to 350 kJ mol^{-1} .
 - 4 Each C atom has a valency of 3.
- A 1, 3 and 4 only
 B 2, 3 and 4 only
 C 1 and 2 only
 D 1 and 3 only

- 20 Ozone depletion potential (ODP) is a measure of the effectiveness of chlorofluoroalkanes in destroying stratospheric ozone.

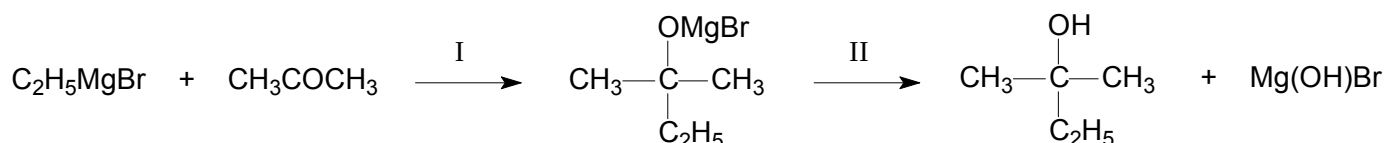
In which sequence are compounds listed in increasing order of their ODPs?

- A $\text{CHClF}_2 < \text{CH}_3\text{CCl}_2\text{F} < \text{CCl}_2\text{FCClF}_2$
 B $\text{CHClF}_2 < \text{CCl}_2\text{FCClF}_2 < \text{CH}_3\text{CCl}_2\text{F}$
 C $\text{CCl}_2\text{FCClF}_2 < \text{CHClF}_2 < \text{CH}_3\text{CCl}_2\text{F}$
 D $\text{CH}_3\text{CCl}_2\text{F} < \text{CCl}_2\text{FCClF}_2 < \text{CHClF}_2$

The following information is relevant to questions **21** and **22**.

Most of the chemistry of magnesium relates to its ionic compounds. However, magnesium does form an important group of covalent compounds which are known as Grignard reagents. Many Grignard reagents, with different alkyl or aryl groups, are widely used in organic syntheses.

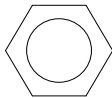
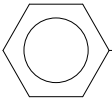
The following reaction scheme shows a typical example of the use of a Grignard reagent.



21 What are the types of reaction shown in this reaction scheme?

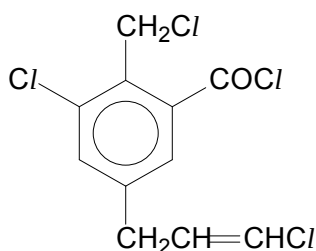
	I	II
A	electrophilic addition	acid–base
B	nucleophilic substitution	elimination
C	nucleophilic addition	acid–base
D	nucleophilic addition	hydrolysis

22 Which of the following sets show the correct reactants and products obtained upon undergoing the same reaction scheme?

	reactants	products
1	 -MgBr + CH ₃ CHO	 - $\overset{\text{OH}}{\underset{\text{H}}{\text{C}}}$ -CH ₃ + Mg(OH)Br
2	CH ₃ CH ₂ CH ₂ MgBr + CH ₃ COCH ₃	CH ₃ CH ₂ - $\overset{\text{OH}}{\underset{\text{CH}_3}{\text{C}}}$ -CH ₃ + Mg(OH)Br
3	CH ₃ CH ₂ CH ₂ MgBr + CO ₂	CH ₃ CH ₂ CH ₂ - $\overset{\text{O}}{\parallel}$ C-OH + Mg(OH)Br

- A** 2 only
B 3 only
C 1 and 2 only
D 1 and 3 only

23 Compound **T** has the structure as shown below.

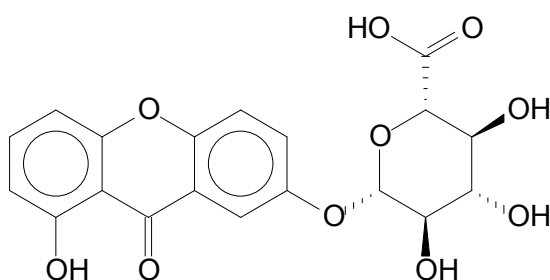


1 mol of compound **T** is warmed with aqueous sodium hydroxide. The resulting solution is cooled and acidified with dilute nitric acid. Excess aqueous silver nitrate is then added to it.

How many moles of silver chloride is precipitated out?

- A** 1 **B** 2 **C** 3 **D** 4

24 The pigment *Indian Yellow*, which first appeared in the 14th century and then vanished at the end of the Victorian era, was made famous by artist Joseph Mallord William Turner.



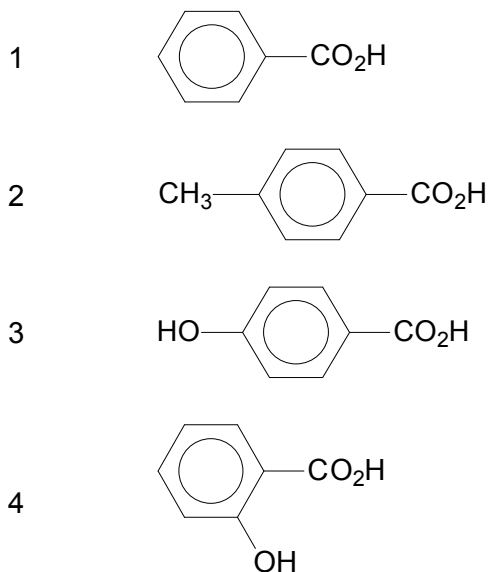
Indian Yellow

In the following reactions, R–O–R remains unaltered.

Which of the following observation is correct?

- A** It reacts with hot, alkaline aqueous iodine to give a yellow solid.
B It reacts with hot, acidified $K_2Cr_2O_7$ to give a compound with 3 carbonyl groups.
C It reacts with aqueous bromine to incorporate up to 6 atoms of bromine in each molecule.
D It reacts with excess sodium metal to produce 2.5 mol of hydrogen gas for each mole of the molecule.

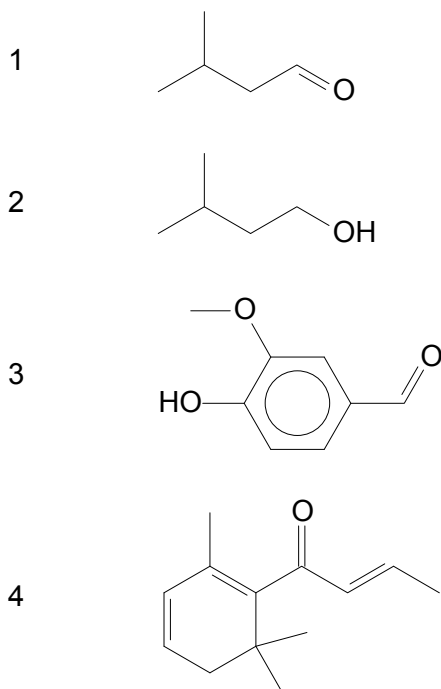
25 Consider the following four carboxylic acids.



What is the relative order of **decreasing** pK_a ?

- | | | | | |
|----------|---|---|---|---|
| A | 3 | 2 | 1 | 4 |
| B | 1 | 2 | 3 | 4 |
| C | 4 | 1 | 2 | 3 |
| D | 2 | 1 | 4 | 3 |

26 The following compounds are responsible for the smell of tequila.

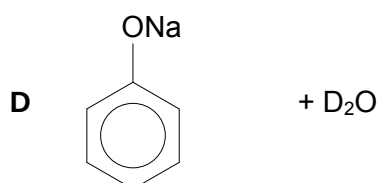
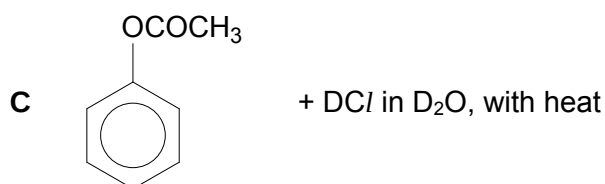
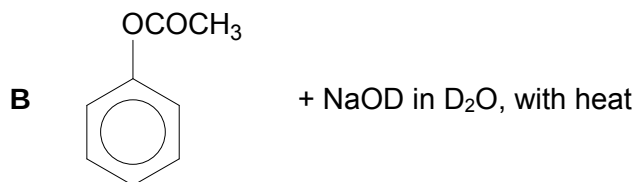
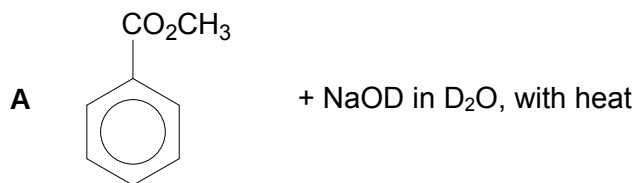


Which of the compounds can be oxidised by acidified potassium dichromate(VI) and can be reduced by sodium borohydride?

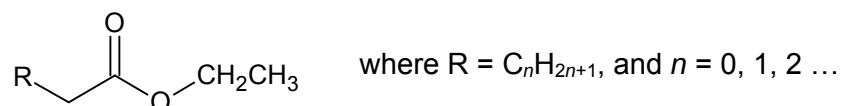
- A 2 only
- B 1 and 3 only
- C 2 and 4 only
- D 1, 3 and 4 only

27 Deuterium, D, is an isotope of hydrogen.

Which compounds will react to produce deuterophenol, C_6H_5OD , in good yield?

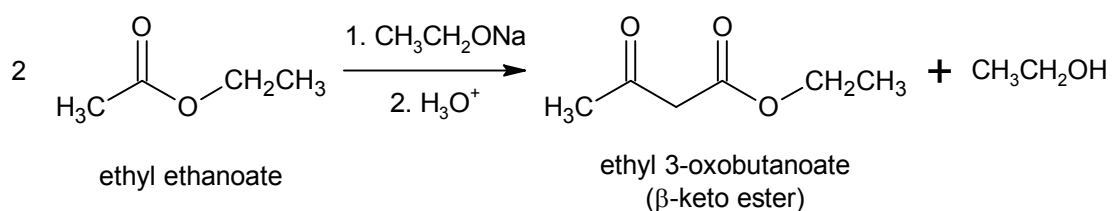


28 Members of an ethyl ester homologous series have the general formula

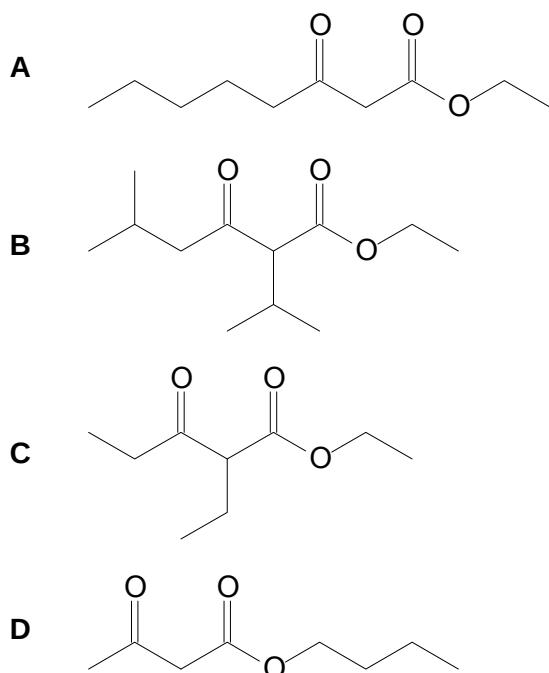


Each member undergoes Claisen condensation with either itself or another member of the series to form a β -keto ester.

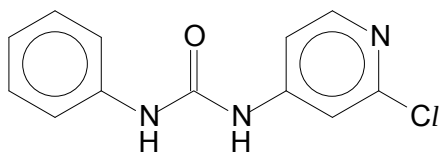
For example, the first member of the series, ethyl ethanoate, combines with itself in the presence of sodium ethanoate, followed by acidification, to form ethyl 3-oxobutanoate. Ethanol is eliminated in the process.



Which of the following is a possible product of the Claisen condensation between ethyl ethanoate and the **fourth** member of the series?

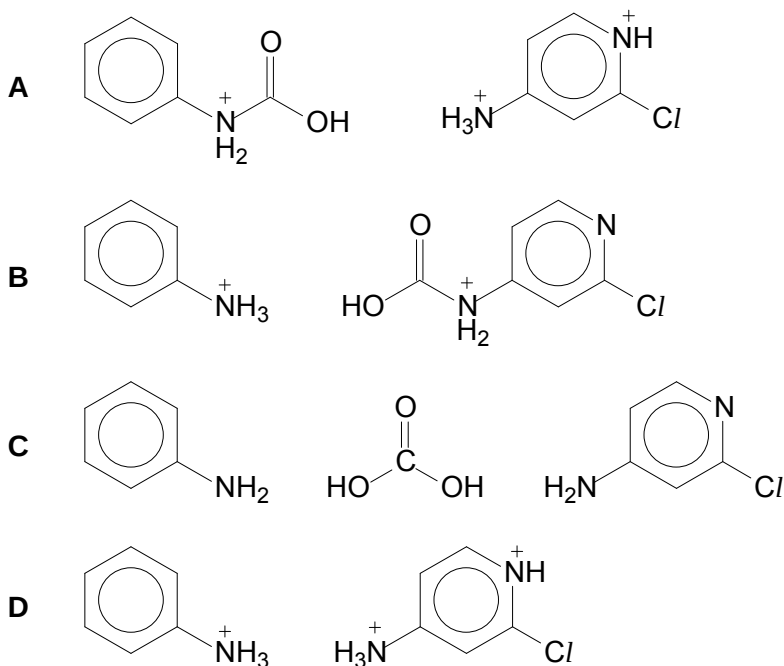


- 29 In 2011, farmers in Eastern China were hit by a spate of exploding watermelons. This was a result of the overuse of *forchlorfenuron*, a plant growth regulator.



forchlorfenuron

What are the organic products formed when forchlorfenuron is heated with dilute acid?



- 30 When excess of NaOH(aq) are added to $\text{Fe(NO}_3)_2\text{(aq)}$, a dirty green precipitate is formed. On standing in air, the precipitate turns reddish brown.

Which process does **not** occur in this sequence?

- A Acid–base
- B Ligand exchange
- C Oxidation
- D Precipitation

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	16	C
2	D	17	B
3	A	18	C
4	D	19	C
5	A	20	A
6	B	21	D
7	C	22	D
8	A	23	B
9	A	24	D
10	C	25	A
11	B	26	B
12	A	27	C
13	C	28	B
14	C	29	D
15	B	30	B

A	7
B	8
C	8
D	7